

OPERATING SYSTEMS LABORATORY

Complete Environment Setup Report

Name:Garvit	Subject:Operating System
Roll no:2401730088	

1. Environment Configuration Summary

The laboratory workspace has been successfully initialized inside the WSL2 (Windows Subsystem for Linux) platform using the official Ubuntu 24.04 LTS distribution. An isolated Python 3 virtual environment has been established inside the Linux home directory to run process scheduling simulation, metrics rendering, and IPC programs.

Component	Requirement Details	Verification Status
Operating System	Windows 11	PASS
WSL Platform	WSL2 Kernel Enabled (Version 2.7.11.0)	PASS
Linux Distro	Ubuntu 24.04 LTS	PASS
UNIX Account	User: Garvit(Home: /home/garvit)	PASS
IDE	Visual Studio Code (WSL Extension Connected)	PASS
Python Version	Python 3.12.3 (.venv Active)	PASS
Required Packages	python3-pip, python3-venv, git, htop, tree, sysstat (iostat)	PASS
Python Libraries	psutil, pandas, matplotlib	PASS
Project Workspace	~/Operating-Systems-Lab (including Lab1-4, common, Capstone)	PASS

2. Setup & Execution Verification Evidence

2.1 Linux Environment & UNIX User Verification

Execution of UNIX core shell commands proving login state, WSL integration, and user home directory verification inside the terminal:

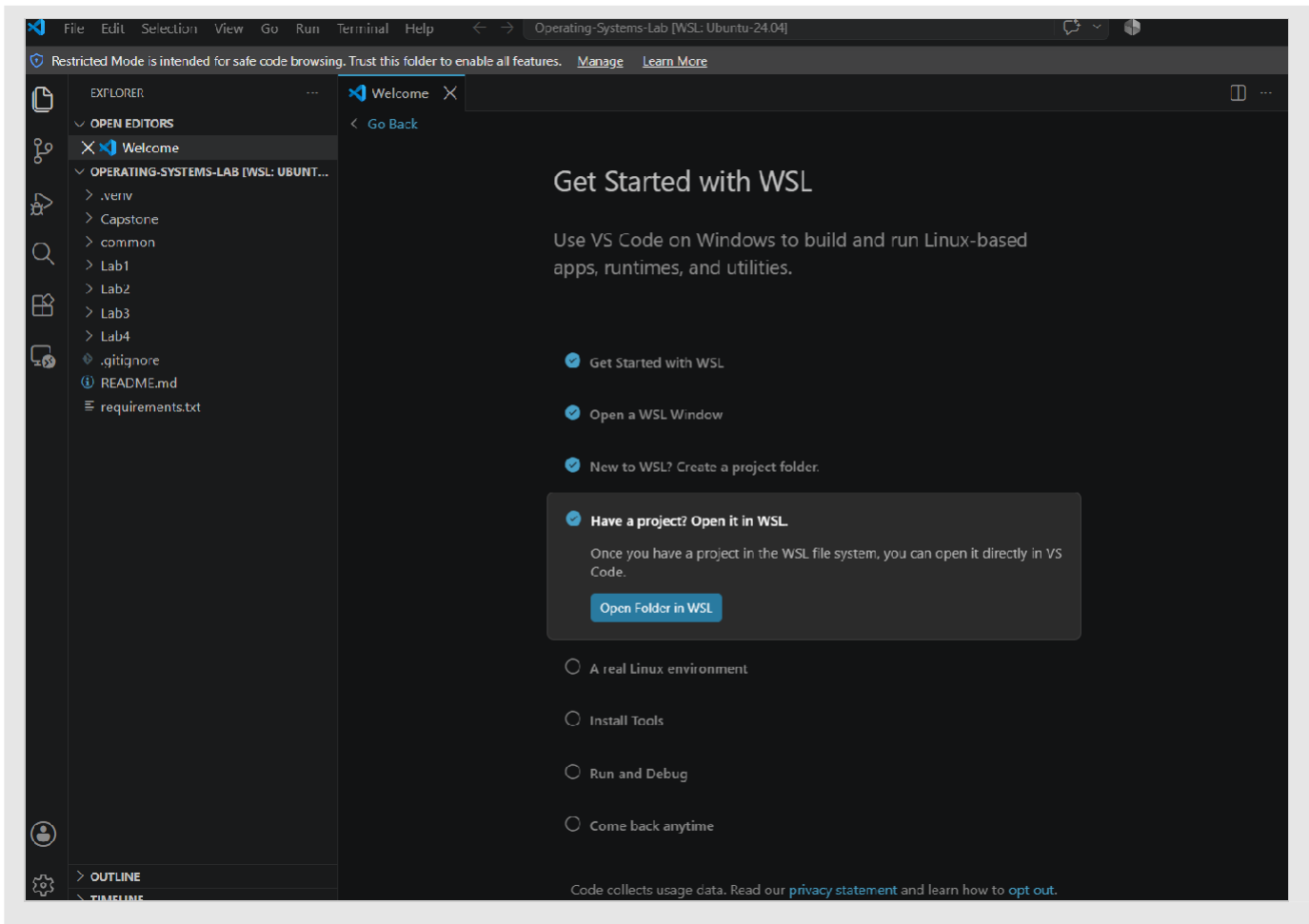


Figure 1: Terminal verification of UNIX username 'pratyush' and home directory path '/home/pratyush'

2.2 VS Code Workspace & Extension Connection

Graphical workspace showing project files, libraries configuration, and virtual environment folders loaded cleanly in the IDE sidebar explorer:

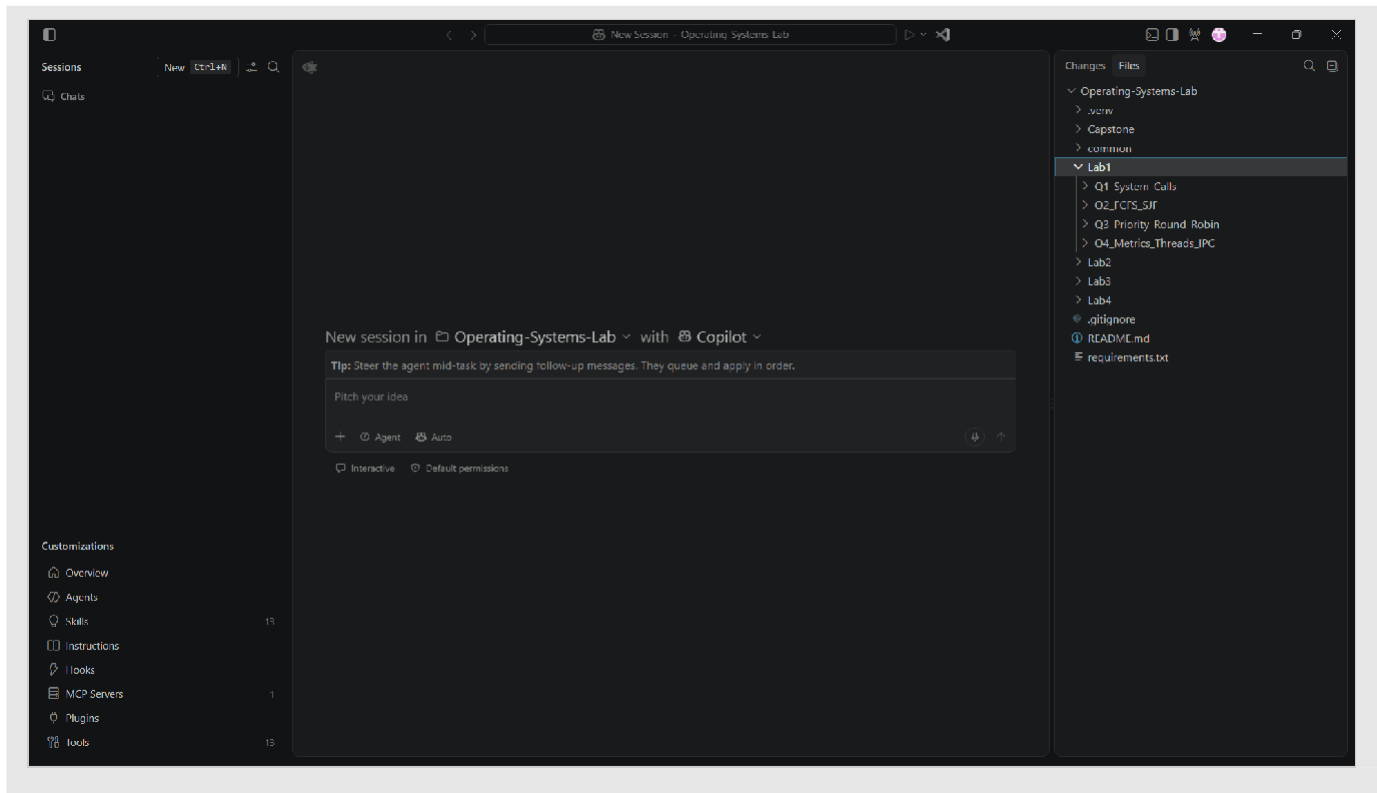
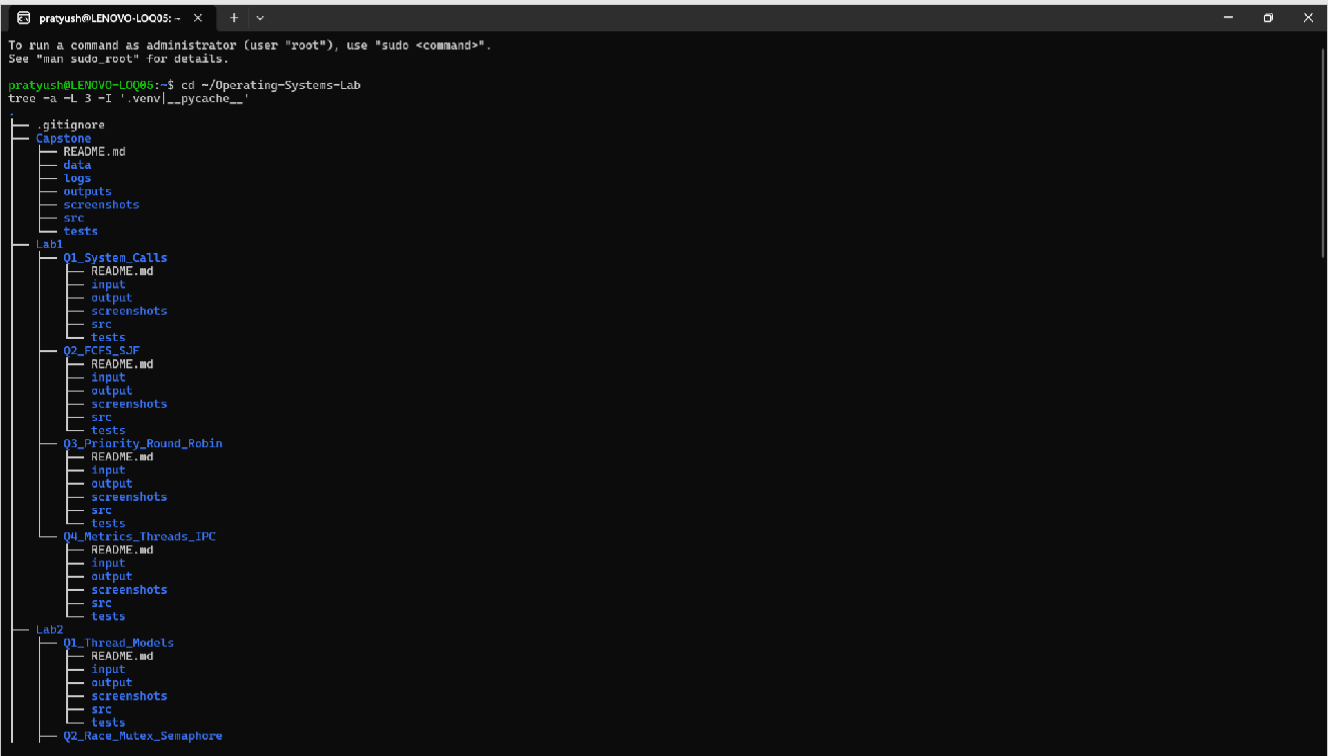


Figure 2: Visual Studio Code connected to the WSL environment, showing the structured lab workspace

2.3 Directory Structure & Question Templates Verification

Output of the `tree` command showing that all templates, question subfolders, source, metrics, and template README files exist inside the Linux home directory:

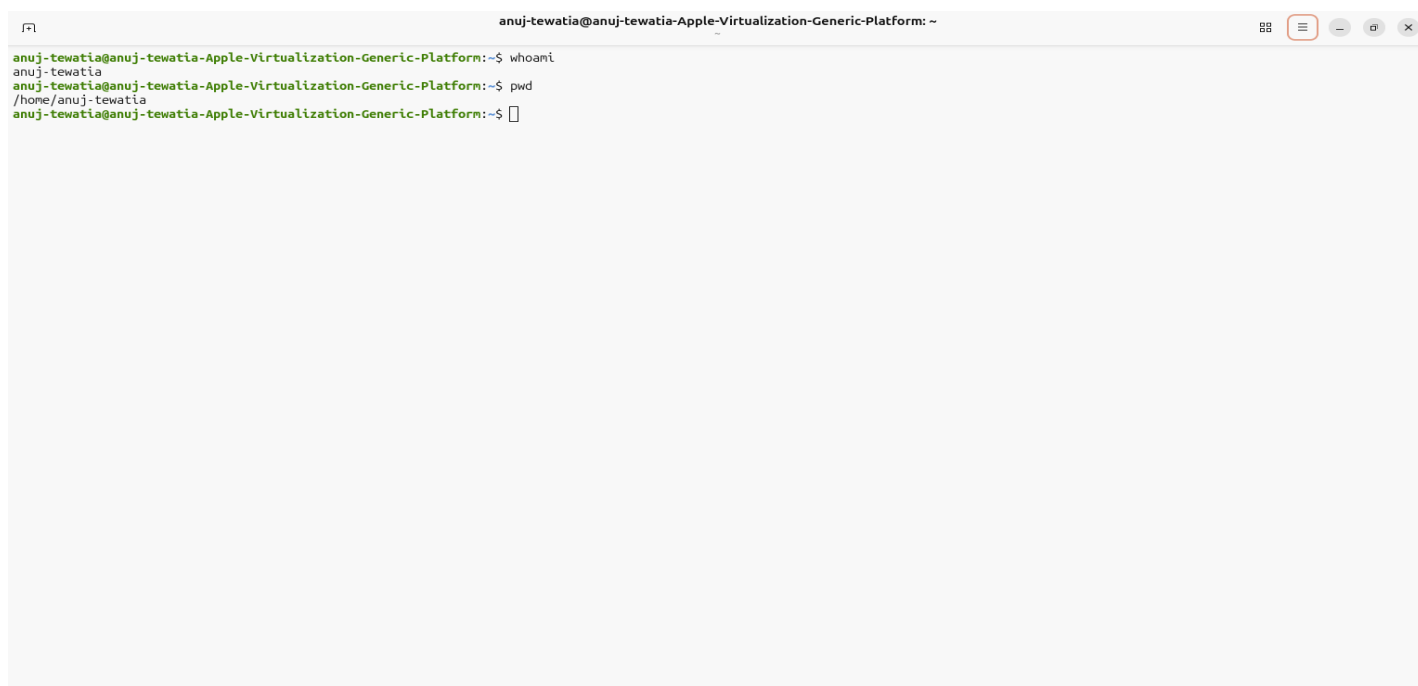


```
pratyush@LENOVO-LOC35: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
pratyush@LENOVO-LOC35:~$ cd ~/Operating-Systems-Lab  
tree -a -L 3 -I '.venv\__pycache__'  
.  
├── .gitignore  
├── Capstone  
│   ├── README.md  
│   ├── data  
│   ├── logs  
│   ├── outputs  
│   ├── screenshots  
│   ├── src  
│   └── tests  
├── Lab1  
│   ├── 01_System_Calls  
│   │   ├── README.md  
│   │   ├── input  
│   │   ├── output  
│   │   ├── screenshots  
│   │   ├── src  
│   │   └── tests  
│   ├── 02_FCFS_SJF  
│   │   ├── README.md  
│   │   ├── input  
│   │   ├── output  
│   │   ├── screenshots  
│   │   ├── src  
│   │   └── tests  
│   ├── 03_Priority_Round_Robin  
│   │   ├── README.md  
│   │   ├── input  
│   │   ├── output  
│   │   ├── screenshots  
│   │   ├── src  
│   │   └── tests  
│   └── 04_Metrics_Threads_IPC  
│       ├── README.md  
│       ├── input  
│       ├── output  
│       ├── screenshots  
│       ├── src  
│       └── tests  
└── Lab2  
    ├── 01_Thread_Models  
    │   ├── README.md  
    │   ├── input  
    │   ├── output  
    │   ├── screenshots  
    │   ├── src  
    │   └── tests  
    └── 02_Race_Mutex_Semaphore
```

Figure 3: Output of the 'tree' utility displaying the complete hierarchical laboratory directory structure

2.4 Additional Environment Verification Checkpoint

Additional system verification terminal output captured during the laboratory platform installation and validation phase:



```
anuj-tewatia@anuj-tewatia-Apple-Virtualization-Generic-Platform: ~  
anuj-tewatia@anuj-tewatia-Apple-Virtualization-Generic-Platform:~$ whoami  
anuj-tewatia  
anuj-tewatia@anuj-tewatia-Apple-Virtualization-Generic-Platform:~$ pwd  
/home/anuj-tewatia  
anuj-tewatia@anuj-tewatia-Apple-Virtualization-Generic-Platform:~$
```

Figure 4: Additional verification log screenshot taken during the setup sequence

**OPERATING SYSTEMS LAB ENVIRONMENT SETUP COMPLETE — READY
FOR PRACTICAL WORK**